BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: PR XII (002) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Physics Paper–XII (A-002) (21)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. A one microfarad of a TV is subjected to 4000 V potential difference. The energy stored in capacitor is.....

(A) 8 J
(B) 16 J
(C) 4×10^{-3} J
(D) 2×10^{-3} J

Q2. Which one of the given are ohmic materials?

(A) Semiconductor diode
(B) Tungsten filament bulb
(C) Thermistor
(D) Metals

Q3. A magnetic field exerts a force of 10 N on a current carrying conductor of length 0.5 m. If 5.0 A current flows in a conductor (L ⊥ B) then the strength of field is.....

(A) 10 T
(B) 5.0 T
(C) 4.0 T
(D) 8.0 T

Q4. When a wire is stretched and its radius becomes r/2, then its resistance will be:

(A) 16 R
(B) 4R
(C) 2R
(D) 0

Q5. Which one of the given device is always connected in series to a P.D. source?

(A) Potentiometer
(B) Ammeter
(C) Voltmeter
(D) Both A & B

Q6. The peak value of sinusoidal voltage in AC circuit is 50 V. The rms value of voltage is roughly equal to.....

(A) 70 V
(B) 40 V
(C) 35 V
(D) 45 V

Q7. The angular momentum of electron in the 3rd orbit of H-atom is.....

(A) h/2π
(B) 2h/π
(C) 3h/2π
(D) 2h/3π

Q8. A 96 gm sample of radioactive nuclide is placed in a container. After 12 minutes only 6 gm of sample is left undecayed. What is the Half-life of radioactive nuclide.

(A) 48 hr
(B) 3 hr
(C) 48 min
(D) 3 min

Q9. The energy of electron in the excited state n = 4 is.....

(A) −13.6 eV
(B) −3.4 eV
(C) −0.85 eV
(D) −1.5 eV

Q10. The device which converts A.C into D.C is called.....

(A) Diode
(B) Transformer
(C) Amplifier
(D) N-type substance

Q11. The half-life of a radioactive material is.....

(A) T_{1/2} = λ/log2
(B) T_{1/2} = 1/λ
(C) T_{1/2} = log2/λ
(D) T_{1/2} = 3λ/log2

Q12. If the K.E of free electron doubles, its de Broglie wavelength changes by the factor.

(A) √2
(B) 1/√2
(C) 2
(D) 1/2

Q13. The positron has charge which is in magnitude equal to the charge on.....

(A) Electron
(B) Proton
(C) β-particle
(D) All these

Q14. A current in a primary coil changes from 5.0 A to 6.0 A in 6.0 sec an induced emf of 10.0 V is produced in the nearby placed secondary coil. The co-efficient of mutual induction is.....

(A) 15 H
(B) 30 H
(C) 50 H
(D) 60 H

Q15. In the reaction _zX^A → _{z−1}X^A + _0Y^1 neutrino + Q Y stands for:

(A) Electron
(B) Positron
(C) Neutron
(D) Anti-neutrino

Q16. Most of the electrons in the base of NPN transistor flow:

(A) Out of the base lead
(B) Into the collector
(C) Into the emitter
(D) Into the base supply

Q17. Which one of the following materials is weakly attracted by magnet?

(A) Ferromagnetic
(B) Diamagnetic
(C) Paramagnetic
(D) All these

Q18. Let a charge of 1 nano-coulomb lies at the centre of a surface, the number of electric field lines leaving the surface are:

(A) 1000
(B) 250
(C) 155
(D) 113

Section-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Attempt any TEN parts. Each part carries FOUR marks.

1. Describe the process of charging a capacitor.

2. Discuss Ohm's Law.

3. The electric potential is constant through a given region of space. Is the electric field zero or non-zero in this region?

4. State Ampere's Law and derive an expression for the magnetic field around a current carrying wire.

5. Show that the relationship $\varepsilon = -\Delta\Phi/\Delta t$ is dimensionally correct.

6. Explain the phenomenon of self-induction. Define coefficient of self induction.

7. How does doubling the frequency affect the reactance of: (a) an inductor (b) a capacitor

8. Differentiate between paramagnetic, diamagnetic and ferromagnetic materials with suitable examples.

9. Explain why in a transistor: (a) the base is thin lightly doped and (b) the collector is large in size

10. If an electron and a proton have the same de Broglie wavelength, which particle has greater speed?

11. The rest mass of a proton is 1.673×10^{-27} kg. At what speed would the mass of the proton be tripled?

12. What are Bohr's postulates about H-atom? Drive an expression for the radii of electron orbit.

13. Explain the term mass defect and binding energy related to a nucleus.

Section-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [5] Describe the construction of capacitor and derive the energy stored in a capacitor.

(b) [4] Each second 10^{18} electrons flow from right to left across a cross-section of a wire attached to the two terminals. Calculate the current in the wire.

QIV.

(a) [5] What is Galvanometer and how it is converted into Ammeter and voltmeter.

(b) [4] A wire carrying 5A current and has length of 10 cm between the poles of a magnet is kept at an angle of 60° to the uniform field of 0.6 T. Find the force acting on the wire.

QV.

(a) [5] What are the main features of photoelectric effect? Discuss the failure of classical physics and success of photon concept in explaining this effect.

(b) [4] Calcualte the longest wave length of radiations for Paschen series.

QVI. Write short note on any two of the following.

(a) [4.5] A.C Through Pure Inductor

(b) [4.5] Radioactivity

(c) [4.5] Full wave rectifier